

SMART DEVICE PROGRAMMING – MODULE 1

Android – Question Bank / Model Question Paper

Module	1 – Android	Total Pages	25 (Source PDF)
Level	Diploma / Polytechnic	Question Type	Theory & Practical

Part A – Very Short Answer Questions

1. What is Android?
2. Who led the development of Android?
3. What is APK? What is it used for?
4. Name any four devices on which variants of Android are used.
5. What was the first commercially available smartphone running Android?
6. What is the purpose of the Android Software Development Kit (SDK)?
7. What does ADT stand for?
8. List any five features of Android.
9. What is an Android Virtual Device (AVD)?
10. What does API stand for?
11. What is the purpose of the AndroidManifest.xml file?
12. Name the main folders/components found in an Android Studio project.
13. What is Dalvik Virtual Machine (DVM)?
14. What is the role of the .dex file in Android?
15. What is the purpose of the res/layout folder?
16. What is LTE?
17. What is GSM?
18. What is CDMA?
19. What is EDGE?
20. Name the official IDE designed specifically for Android development.

Part B – Short Answer Questions

1. Explain the important milestones in the history of Android from 2003 to 2016.
2. Write the Android versions from Cupcake to Oreo along with their version numbers and code names.
3. Explain any five features of Android.
4. Describe the five major layers of Android architecture.
5. Explain the Android Runtime and the role of Dalvik Virtual Machine.
6. Explain the advantages of DVM mentioned in the module.
7. List the different types of Android devices available in the market.

8. Explain the Android SDK and list the tools/resources it provides.
9. Explain Android Development Tools (ADT) and its functions.
10. List the various mobile technologies mentioned in the module.
11. Compare Android and iOS with respect to developer, release, customizability, source model and file transfer.
12. Explain Eclipse IDE and list its features.
13. Explain Android Studio IDE and mention its important features.
14. Explain the Visual Layout Editor, Fast Emulator and Intelligence Code Editor in Android Studio.
15. What is AVD? Explain why AVDs are useful for Android application testing.
16. Explain the steps for creating an Android Virtual Device.
17. Explain the steps for creating an Android Studio project.
18. Explain the basic components of the Hello World Android program.
19. Explain API Level in Android.
20. Explain the purpose of the Manifests, Java, res and Gradle Scripts folders.

Part C – Long Answer / Essay Questions

1. Explain Android architecture in detail. Describe the Linux Kernel, Libraries, Android Runtime, Application Framework and Applications layers.
2. Describe the history and evolution of Android. Include the major events from the founding of Android Inc. to the introduction of Pixel and Pixel XL.
3. Explain Android versions from Android 1.0 to Android 8.1, including year, version number, code name and API level.
4. Explain Android Studio in detail. Discuss its features such as optimized development environment, Instant App Run, Visual Layout Editor, Fast Emulator, intelligent code editor, activity templates, Firebase connectivity, Kotlin support and colour previews.
5. Explain the complete procedure for creating an Android Virtual Device (AVD), starting from opening AVD Manager through selecting hardware and system image to completing the AVD.
6. Explain the complete procedure for creating an Android Studio project, including project configuration, form factor/API selection, activity selection and activity configuration.
7. Explain the Android communication technologies 1G, 2G, 3G, 4G and 5G, including their important characteristics and data rates given in the module.
8. Explain CDMA, EDGE, LTE and GSM. Include their main characteristics and applications as described in the module.
9. Explain the structure of an Android Studio project. Discuss AndroidManifest.xml, Java/Kotlin source files, drawable, layout, resource and Gradle Scripts folders.
10. Write and explain the Hello World Android program using MainActivity.java, activity_main.xml and AndroidManifest.xml.

Part D – Compare / Differentiate

1. Differentiate between Android and iOS.
2. Differentiate between 1G, 2G, 3G, 4G and 5G.
3. Differentiate between GSM and CDMA.
4. Differentiate between Eclipse IDE and Android Studio IDE.
5. Differentiate between Android SDK and ADT.
6. Differentiate between an actual Android device and an Android Virtual Device (AVD).

Part E – Practical / Procedure-Based Questions

1. Write the steps to install Android Studio on Windows.
2. Write the steps to install the Android SDK in Android Studio.
3. Write the steps to create an AVD.
4. Write the steps to create a new Android Studio project.
5. Write a basic Hello World Android program using an Activity, TextView and AndroidManifest.xml.
6. Identify the purpose of the following Android Studio project items: AndroidManifest.xml, Java/Kotlin folder, drawable, layout, res and Gradle Scripts.

Note: This question paper is prepared from the supplied “Smart Device Programming 2023 – Module 1: Android” PDF. The questions follow the terminology and topics presented in that source.